

Technology #160

Embed an ANN-driven adaptive droop controller
in the rooftop solar inverter that
continuously modifies droop coefficients
based on ambient temperature and load
conditions to limit over-current and heat
generation

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Embed an ANN■driven adaptive droop controller in the rooftop solar inverter that continuously modifies droop coefficients based on ambient temperature and load conditions to limit over■current and heat generation

WHY DOES IT MATTER?

Documented problem 'thermal degradation' concerning rooftop solar inverter. User-submitted engineering problem (session input): "Why do commercial rooftop solar inverters fail prematurely in hot climates?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Physics-of-Failure (PoF) based reliability assessment for photovoltaic (PV) inverters requires long-duration electrical and electrothermal stress histories, yet generating such stress histories with h'; 'Growth in power requirements and the complexity of emerging distribution systems are creating the need for new solid-state products for many applications. The qualification requirements, testing stand'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in thermal. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Operate two identical inverters in a climate chamber at 45 °C with fluctuating irradiance; one uses fixed droop settings, the other uses.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: temperature-rise mapping at max duty. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden; interface contact resistances dominate at small scale and must be measured, not assumed zero; steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected); TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.; TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.; TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.; Model input variables requiring sourced values before any numeric result: h ($W/(m^2.K)$), A (m^2), dT (K), L_i (m), k_i ($W/(m.K)$), A_i (m^2), k ($W/(m.K)$), A (m^2), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , $W/(m.K)$] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m^2] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Adaptive droop characteristic adjustment via ANN■assisted real■time tuning of droop coefficients and PI gains

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'thermal degradation' concerning rooftop solar inverter. User-submitted engineering problem (session input): "Why do commercial rooftop solar inverters fail prematurely in hot climates?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Physics-of-Failure (PoF) based reliability assessment for photovoltaic (PV) inverters requires long-duration electrical and electrothermal stress histories, yet generating such stress histories with h'; 'Growth in power requirements and the complexity of emerging distribution systems are creating the need for new solid-state products for many applications. The qualification requirements, testing stand'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: thermal

Disciplines: heat transfer, thermal engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Embed an ANN-driven adaptive droop controller in the rooftop solar inverter that continuously modifies droop coefficients based on ambient temperature and load conditions to limit overcurrent and heat generation

Subsystems:

?: heat source characterization - ?

?: conduction path / spreader - ?

?: reject interface - ?

?: safety limit interlock - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Embed an ANN-driven adaptive droop controller in the rooftop solar inverter that continuously modifies droop coefficients based on ambient temperature and load conditions to limit overcurrent and heat generation

Key Physics: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: h , dT .

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k , dT/dx .

Assumptions:

- h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden
- interface contact resistances dominate at small scale and must be measured, not assumed zero
- steady-state conduction (no transient term)
- isotropic, homogeneous medium conductivity k
- one-dimensional gradient (lateral gradients neglected)
- TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.
- TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.
- TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.

- Model input variables requiring sourced values before any numeric result: h (W/(m².K)), A (m²), dT (K), L_i (m), k_i (W/(m.K)), A_i (m²), k (W/(m.K)), A (m²), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , W/(m.K)] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m²] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'thermal degradation', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'hotspot exceeding material/safety limit', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'interface delamination raising resistance over life', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coolant/flow path blockage (if active)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'thermal runaway coupling to electronics', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'condensation at cold surfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain >95% availability at ambient temperatures above 40°C	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Adaptive droop characteristic adjustment via ANN-assisted real-time tuning of droop coefficients and PI gains	MODELED
DI-003	Expected effect target	Lowered semiconductor junction temperatures, enabling >95% availability at ambient 40°C and extending inverter service life	MODELED
DI-004	Domain design input pattern: distribution applicable by system	ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: ambient and operating conditions	[ambient, 'conditions']	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: safety-critical applications	[source, 'temperature']	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance standards	IEEE 690.10 (general electrical safety); UL 1741 (inverter safety)	UNKNOWN (never inferred)
DI-008	Custodied observation: Performance and stability analysis of adaptive droop control	IEEE 690.10 (general electrical safety); UL 1741 (inverter safety)	UNKNOWN (never inferred)
DI-009	Custodied observation: Intelligent energy management	IEEE 690.10 (general electrical safety); UL 1741 (inverter safety)	UNKNOWN (never inferred)
DI-010	Custodied observation: A Review of Floating Photovoltaic Systems	IEEE 690.10 (general electrical safety); UL 1741 (inverter safety)	UNKNOWN (never inferred)
DI-011	Custodied observation: ANN-augmented adaptive droop control	IEEE 690.10 (general electrical safety); UL 1741 (inverter safety)	UNKNOWN (never inferred)
DI-012	Custodied observation: Engineering of smart devices for energy applications	IEEE 690.10 (general electrical safety); UL 1741 (inverter safety)	UNKNOWN (never inferred)

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	heat source characterization realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients	UNKNOWN	heat source characterization realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients
DO-002	conduction path / spreader realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients	UNKNOWN	conduction path / spreader realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients
DO-003	reject interface realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients	UNKNOWN	reject interface realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients
DO-004	safety limit interlock realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients	UNKNOWN	safety limit interlock realizing Embed an ANN-driven adaptive droop controller that continuously modifies droop coefficients
DO-005	heat source characterization interfaces with conduction path / spreader	UNKNOWN	heat source characterization interfaces with conduction path / spreader
DO-006	conduction path / spreader interfaces with reject interface	UNKNOWN	conduction path / spreader interfaces with reject interface
DO-007	reject interface interfaces with safety limit interlock	UNKNOWN	reject interface interfaces with safety limit interlock
DO-008	Device boundaries: patient/fluid inlet and outlet terminal	UNKNOWN	Device boundaries: patient/fluid inlet and outlet terminal
DO-009	Flow/force/signal path geometry realizing the mechanism	UNKNOWN	Flow/force/signal path geometry realizing the mechanism
DO-010	Operating range for heat generation at source within which the mechanism holds its stated behavior	UNKNOWN	Operating range for heat generation at source within which the mechanism holds its stated behavior
DO-011	Operating range for spreader conductivity within which the mechanism holds its stated behavior	UNKNOWN	Operating range for spreader conductivity within which the mechanism holds its stated behavior
DO-012	Operating range for interface contact resistance within which the mechanism holds its stated behavior	UNKNOWN	Operating range for interface contact resistance within which the mechanism holds its stated behavior
DO-013	Operating range for reject-path area within which the mechanism holds its stated behavior	UNKNOWN	Operating range for reject-path area within which the mechanism holds its stated behavior
DO-014	Operating range for allowable temperature rise within which the mechanism holds its stated behavior	UNKNOWN	Operating range for allowable temperature rise within which the mechanism holds its stated behavior
DO-015	Operating range for ambient worst case within which the mechanism holds its stated behavior	UNKNOWN	Operating range for ambient worst case within which the mechanism holds its stated behavior
DO-016	Sensing -> decision -> actuation loop maintaining the proposed operating position	UNKNOWN	Sensing -> decision -> actuation loop maintaining the proposed operating position
DO-017	Information path: sensing element -> decision/monitor	UNKNOWN	Information path: sensing element -> decision/monitor
DO-018	Fixture enabling: temperature-rise mapping at max duty	UNKNOWN	Fixture enabling: temperature-rise mapping at max duty
DO-019	Fixture enabling: interface resistance characterization	UNKNOWN	Fixture enabling: interface resistance characterization
DO-020	Fixture enabling: worst-case ambient soak test — rigorous	UNKNOWN	Fixture enabling: worst-case ambient soak test — rigorous

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: temperature-rise mapping at maximum	UNKNOWN	Unknown concept name; fixture specific instrumentation selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
heat generation at source	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
spreader conductivity	UNKNOWN (no sourced value)	W/(m.K)	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
interface contact resistance	UNKNOWN (no sourced value)	K/W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
reject-path area	UNKNOWN (no sourced value)	m^2	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
allowable temperature rise	UNKNOWN (no sourced value)	K	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
ambient worst case	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
thermal degradation	Fails by thermal degradation design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions
hotspot exceeding material safety limit	Fails by hotspot exceeding material safety limit; heat-seal degradation; overheating; catastrophic failure; material degradation	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions
interface delamination	Fails by interface delamination; loss of electrical contact; overheating; catastrophic failure; material degradation	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions
coolant/flow path blockage	Fails by coolant/flow path blockage (if active); particulate debris blockage; overheating; catastrophic failure; material degradation	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions
thermal runaway coupling	Fails by thermal runaway coupling; overheating; catastrophic failure; material degradation	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions
condensation at cold surfaces	Fails by condensation at cold surfaces; overheating; catastrophic failure; material degradation	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions	design occurs in the STA at 600°C (exceeds design limit) when operated under the proposed conditions

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
thermal degradation	Fails by thermal degradation: the device design that on NOA ESTABLISHED (to be fixed after verification)	design that on NOA ESTABLISHED (to be fixed after verification)	design that on NOA ESTABLISHED (to be fixed after verification)
hotspot exceeding material safety limit	Fails by hotspot exceeding material/safety limit	design that on NOA ESTABLISHED (to be fixed after verification)	design that on NOA ESTABLISHED (to be fixed after verification)
interface delamination	Fails by interface delamination	design that on NOA ESTABLISHED (to be fixed after verification)	design that on NOA ESTABLISHED (to be fixed after verification)
coolant/flow path blockage	Fails by coolant/flow path blockage (if active)	design that on NOA ESTABLISHED (to be fixed after verification)	design that on NOA ESTABLISHED (to be fixed after verification)
thermal runaway coupling	Fails by thermal runaway coupling to electronics	design that on NOA ESTABLISHED (to be fixed after verification)	design that on NOA ESTABLISHED (to be fixed after verification)
condensation at cold surfaces	Fails by condensation at cold surfaces: surface is below the operating point	design that on NOA ESTABLISHED (to be fixed after verification)	design that on NOA ESTABLISHED (to be fixed after verification)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Operate two identical inverters in a climate chamber	Method: climate chamber	Acceptance: criteria to be fixed	Result: criteria to be fixed
VF-002	run killer experiment: falsification_test	Method: falsification_test	Acceptance: criteria to be fixed	Result: criteria to be fixed
VF-003	temperature-rise mapping at max duty	Method: temperature-rise mapping	Acceptance: criteria to be fixed	Result: criteria to be fixed
VF-004	temperature-rise mapping at maximum	Method: temperature-rise mapping	Acceptance: criteria to be fixed	Result: criteria to be fixed
VF-005	thermal-resistance trend over acceleration	Method: thermal-resistance trend	Acceptance: criteria to be fixed	Result: criteria to be fixed
VF-006	temperature-rise mapping at max duty	Method: temperature-rise mapping	Acceptance: criteria to be fixed	Result: criteria to be fixed
VF-007	worst-case ambient soak with staged power	Method: worst-case ambient soak	Acceptance: criteria to be fixed	Result: criteria to be fixed

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	Method: end-use validation	Acceptance: criteria to be fixed	Result: criteria to be fixed

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible thermal spreader (graphite/AlN)	graphite/AlN	electronics thermal practice	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible thermal spreader	biocompatible thermal spreader (graphite/AlN)	1	STANDARD	biocompatible thermal spreader (graphite/AlN)	Not Established	STANDARD	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- bonded spreader + encapsulation: domain registry manufacturing route - candidate only; no process qualification exists
- thermal interface material application control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter heat generation at source [q W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter spreader conductivity [k_i W/(m.K)]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter interface contact resistance [R_total K/W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter reject-path area [A m^2]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter allowable temperature rise [dT K]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter ambient worst case [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification	physical verification (0.5 GHz) static power	NOT ESTABLISHED	initial hardware	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope	per method	constraint-compliance	NOT ESTABLISHED	constraint compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Operate two identical	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	temperature-rise map	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	temperature-rise map	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	thermal-resistance trend	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	temperature-rise map	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	worst-case ambient	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.